Cleveland Heart Disease Exploratory Data Analysis

Maximum Exercise Heart Rate and Recorded Heart-Disease Status

Data Science Final Portfolio Report

Prepared by: Ramisha Tasfia

Dataset: UCI Heart Disease Cleveland subset

Analysis: 303 historical clinical observations

Executive summary

The purpose of this study was to compare the difference between the maximum exercise heart rate of patients with and without documented heart disease in the UCI Cleveland dataset. For the primary comparison, all 303 observations were retained. Patients without recorded disease (n = 164) had a mean maximum HR of 158.38 beats per minute (bpm) while patients with recorded disease (n = 139) had a mean maximum HR of 139.26 bpm. The unadjusted mean difference was 19.12 bpm (95% CI: 14.33–23.91). Welch's two-sided t-test found evidence of a group difference ($t \approx 7.858$, $p \approx 9.11 \times 10^{-14}$). These are association not causal or individual diagnostic results.

1. Research question and hypotheses

Hypothesis: Heart disease patients will have a lower maximum heart rate during exercise (thalach, bpm) than non-heart disease patients?

The null hypothesis (H_0): There is no difference between the population means of maximum exercise heart rate for the two recorded-disease groups. Alternative hypothesis (H_1): The population means are different. The main level of significance is $\alpha = 0.05$ with twosided comparison.

2. Dataset and provenance

Repository	UCI Machine Learning Repository
Dataset	Heart Disease processed Cleveland subset
Original contributors	Andras Janosi; William Steinbrunn; Matthias Pfisterer; Robert Detrano
Observations / attributes	303 records / 14 fields
Location and context	Historical Cleveland clinical sample, Ohio, United States
Original target	num: 0 = no recorded disease; 1–4 = recorded disease
Source DOI	10.24432/C52P4X
License	CC BY 4.0 (attribution required)

The analysis CSV file was generated from the original processed.cleveland.data file, with the column names added and the missing value marker (?) interpreted as missing. It's an analysis copy - it is not a byte-for-byte copy of the source file.

3. Data quality and cleaning

Initial records	**303**
Missing entries	6 total: ca = 4; thal = 2
Exact duplicate rows	0
Primary-analysis records retained	303
Heart-rate outliers flagged by 1.5 × IQR	1; retained pending clinical context
Primary comparison treatment	No imputation or row removal required for thalach and binary disease status
Optional model treatment	Training-only median imputation of missing predictors

Before modeling, numeric types, plausible ranges of coding and duplicates are checked. The number of missing values in ca and thal is not important for the comparison of the primary heart rates and those records should not be deleted, as this would be an unnecessary loss of samples. This is because the single IQR-flagged observation of the heart-rate data set is not necessarily an error and it was not discarded. Both the original CSV and a cleaning audit are saved separately.

4. Descriptive findings

Without recorded disease	**n = 164; mean thalach = 158.38 bpm**
With recorded disease	n = 139; mean thalach = 139.26 bpm
Unadjusted mean difference	19.12 bpm (without minus with recorded disease)
Standardized difference	Cohen's d ≈ 0.918

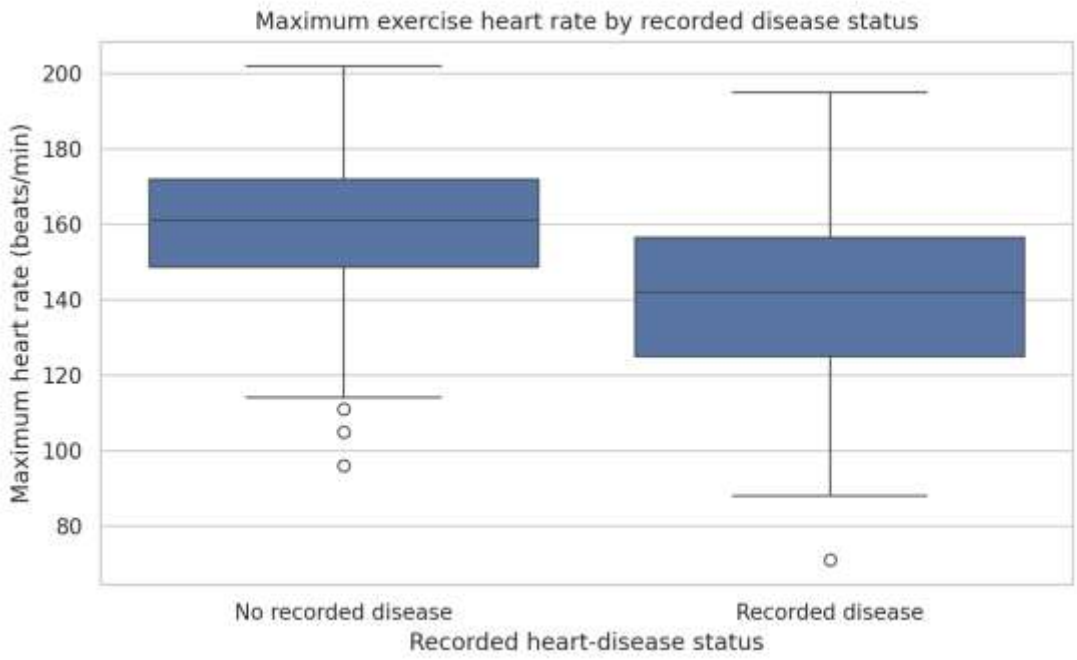


Figure 1. Maximum exercise heart rate by recorded disease status (boxplots).

Records with no disease had a higher level of central heart rate (Figure 1). The significant overlap suggests that no single heart-rate measurement is a good enough discriminant to distinguish between persons.

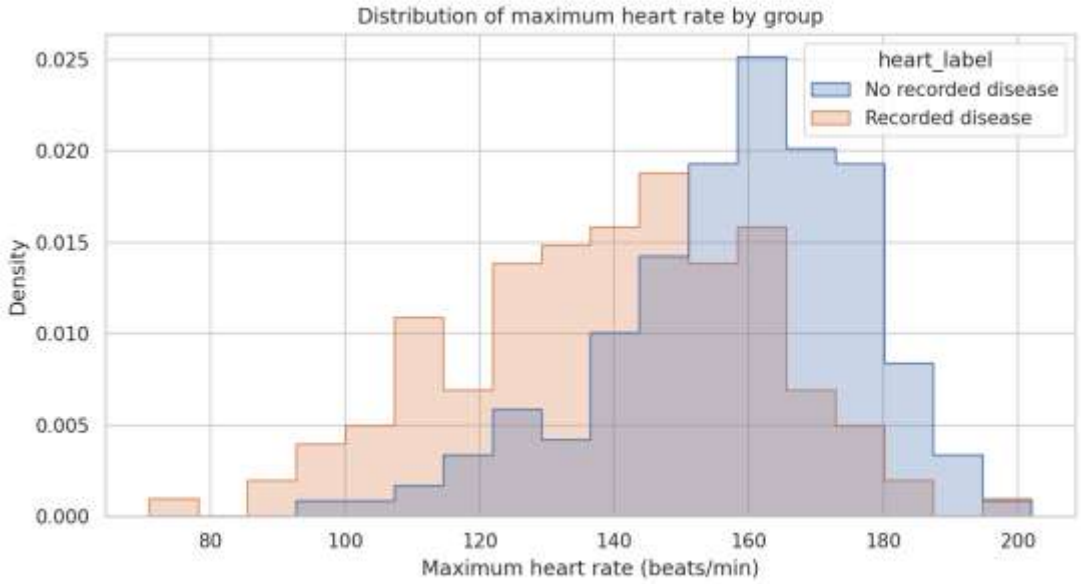


Figure 2. Distribution of maximum exercise heart rate in both groups.

The no-recorded-disease distribution shifted toward higher rates in figure 2, with both groups having a wide range of variation. This indicates that formal inference can be aided, but may offer caution in using a diagnostic threshold.

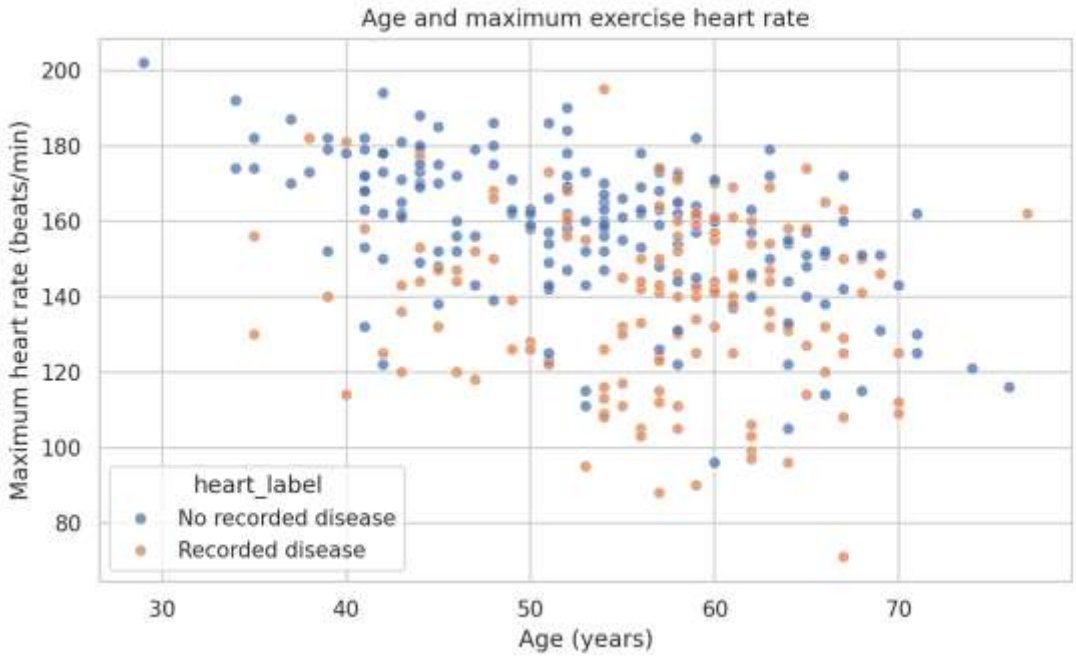


Figure 3. Age and maximum exercise heart rate, grouped by disease status.

Figure 3 shows the reasons why age might be a factor when making an unadjusted group comparison: Age can be correlated with exercise heart rate, and samples can vary in age distribution between disease groups.

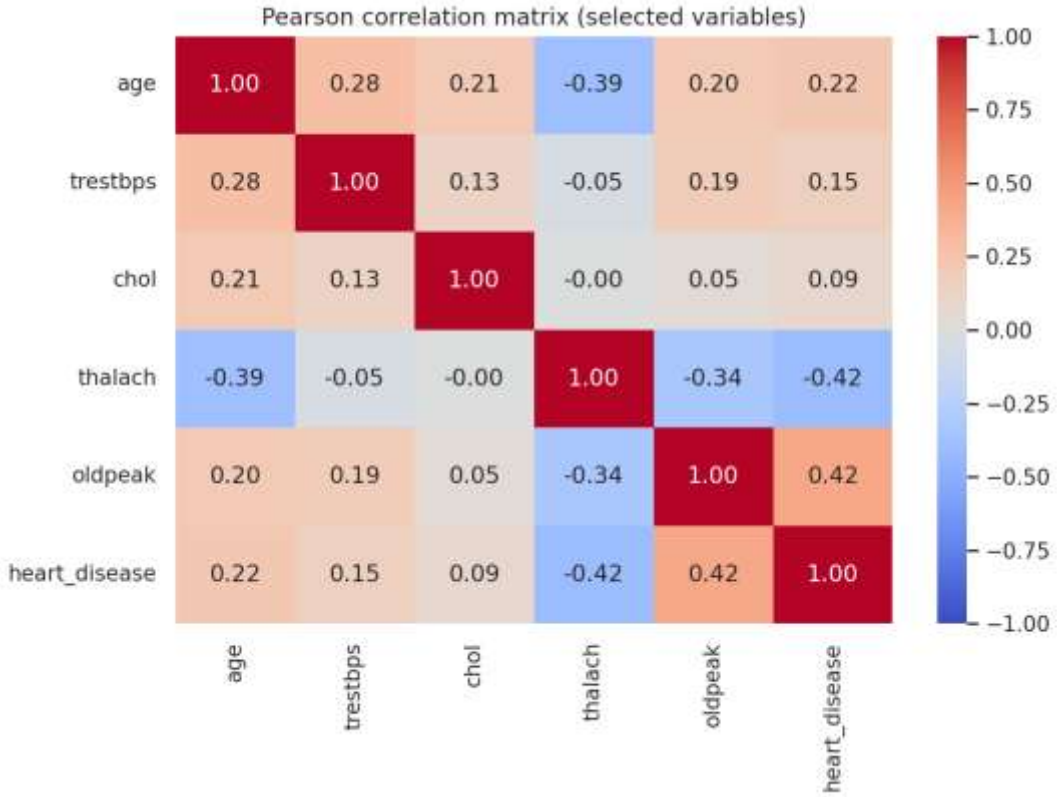


Figure 4. Correlation heatmap of selected numeric variables.

Figure 4 is exploratory; this means that the correlations are linear associations, there are no controls, and causation cannot be inferred. The binary disease indicator is not a disease severity scale.

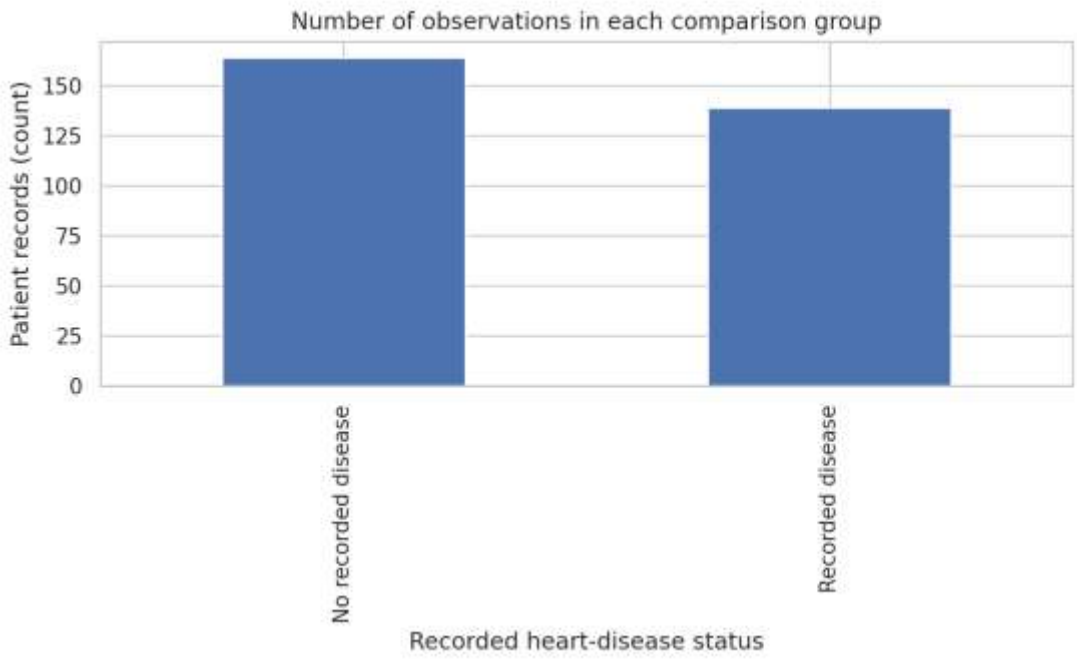


Figure 5. Number of records in each disease-status group.

A total of 164 (without disease) and 139 (with disease) records were obtained from Figure 5. The sample sizes are sufficiently large, but not population representative, for both groups for this exploratory comparison.

5. Statistical analysis

A two-sided Welch independent-samples t-test was used due to the outcome being numeric and the assumption of independence of the two groups, with the assumption of equal variances in the populations. In addition, the notebook performs a Mann–Whitney U sensitivity test, which is not the same as the hypothesis of equal means, and is a rank/distribution comparison.

Test	Welch's two-sided independent-samples t-test
Mean difference	19.12 bpm
95% confidence interval	14.33 to 23.91 bpm
t statistic	≈ 7.858
p-value	$\approx 9.11 \times 10^{-14}$
Decision at α = .05	Reject equal-means null hypothesis
Effect size	Cohen's d ≈ 0.918

The confidence interval measures the uncertainty about the unadjusted population mean difference, given the model assumptions. The p-value does not indicate the probability of the null hypothesis being true, but rather is evidence against equality of the group means. The test is based on the

assumption of independent records, the data alone cannot test for independence or remove confounding from the data.

6. Optional predictive model

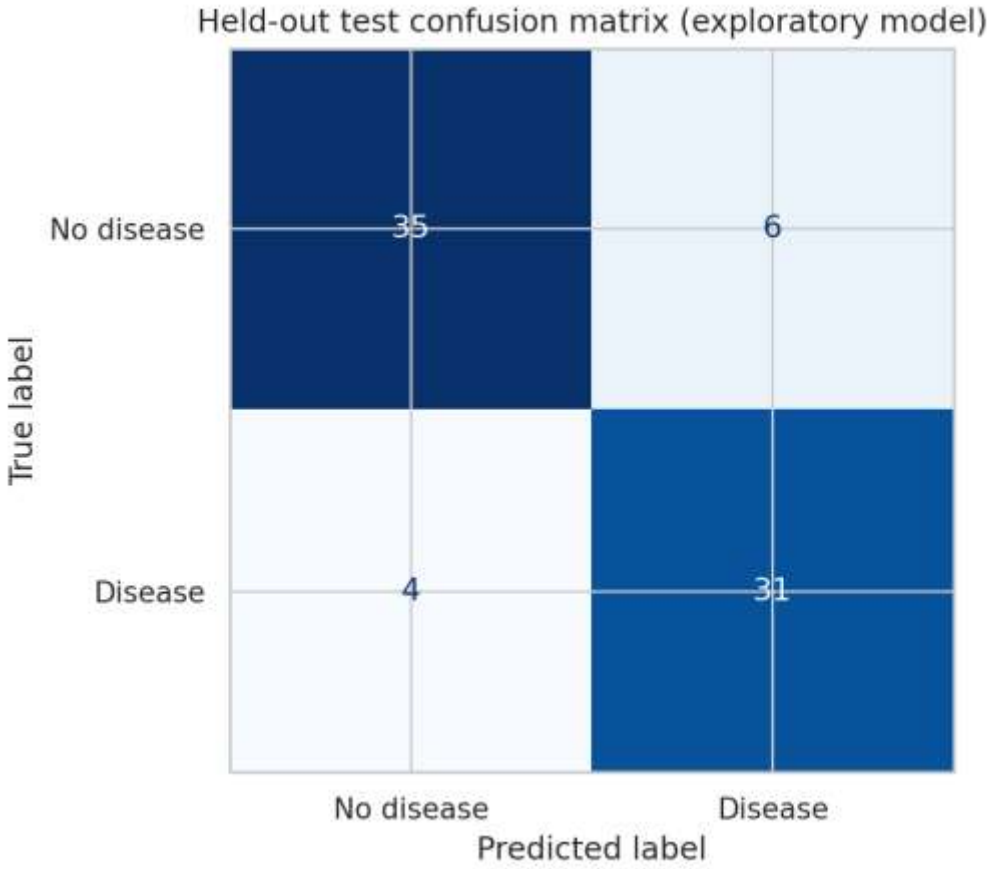


Figure 6. Confusion matrix for the optional held-out classification demonstration.

A logistic-regression classifier was tested on a stratified held-out test set, with no context provided, except for a baseline classifier based on the maximum number of majority classes. The notebook. previously executed recorded around 86.8% test accuracy, 83.8% precision, and 88.6% recall, when contrasted to a 53.9% base line accuracy. These are exploratory measurements based on past data and are not clinically reliable nor valid for external use.

7. Limitations, ethics, and responsible interpretation

This is a small sample of the history of clinical cases, not a random sample of the U.S. population. Original categories 0-4 for diseases were simplified to a binary indicator of disease status, with no severity detail. This difference may be confounded by other factors not measured, such as age, fitness, medication, referral patterns, etc. This analysis does not imply causality and cannot be used for individual patients for diagnosis. These results are for educational purposes only, not medical advice. The distribution of the dataset must be accompanied by an attribution as per CC BY 4.0.

8. Conclusion

There is a statistically significant difference in the maximum HR for records with and without a history of heart disease in Cleveland data. The mean rate was about 19 bpm lower in the recorded-disease group (95% confidence interval not including zero). Although statistical evidence is provided, there is overlap in group distributions and the study does not provide evidence for causation or a clinical screening rule. Before more general conclusions, more representative, adjusted and externally validated work would be required.

9. Reproducibility

The analysis was performed in a Python Jupyter Notebook with Pandas and NumPy to process the data, Matplotlib and Seaborn to visualize results, and SciPy and Scikit-learn to perform statistical analysis and apply machine learning techniques. The dataset is from the UCI Machine Learning Repository and consists of 303 patients from the Cleveland Heart Disease dataset.

The project contains the original CSV file (data/cleveland_heart_disease.csv), the Jupyter Notebook with a full analysis, a data-cleaning log file (cleaning_log.csv), generated visualizations, and a requirements.txt file with the required Python libraries.

The analysis can be repeated by installing all the required dependencies, launching the Jupyter Notebook, and clicking on Restart & Run All. The notebook records details of data cleaning, statistical computations, visualizations and the assessment of the model, so that the results can be independently scrutinized and validated.

10. References and AI-assistance disclosure

10.1 References

Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1988). *Heart Disease* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C52P4X

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10.2 AI-Assistance Disclosure

ChatGPT was used as a supporting tool during this project, including assistance with dataset selection, Python code development, statistical analysis, data visualization, and report preparation.

The Cleveland Heart Disease dataset was analyzed using Python in Jupyter Notebook. The notebook was executed, and its outputs were examined as part of the project workflow. The findings and conclusions are based on the resulting statistical calculations and visualizations.

A concise record of AI-assisted activities and their purposes is provided in the accompanying prompt_log.md file.